how a self-contained universe could have an earliest moment in time, presumably identified with the Big Bang.

While Wheeler and others thought about the technical challenges of quantum gravity, Everett became fascinated by these conceptual problems, especially how to handle measurement. The seeds of the Many-Worlds formulation can be traced to a late-night discussion in 1954 with fellow young physicists Charles Misner (also a student of Wheeler's) and Aage Petersen (an assistant of Bohr's, visiting from Copenhagen). All parties agree that copious amounts of sherry were consumed on the occasion.

Clearly, Everett reasoned, if we're going to talk about the universe in quantum terms, we can't carve out a separate classical realm. Every part of the universe will have to be treated according to the rules of quantum mechanics, including the observers within it. There will only be a single quantum state, described by what Everett called the "universal wave function" (and we've been calling "the wave function of the universe").

If everything is quantum, and the universe is described by a single wave function, how is measurement supposed to occur? It must be, Everett reasoned, when one part of the universe interacts with another part of the universe in some appropriate way. That is something that's going to happen automatically, he noticed, simply due to the evolution of the universal wave function according to the Schrödinger equation. We don't need to invoke any special rules for measurement at all; things bump into each other all the time.

It's for this reason that Everett titled his eventual paper on the subject "'Relative State' Formulation of Quantum Mechanics." As a measurement apparatus interacts with a quantum system, the two become entangled with each other. There are no wave-function collapses or classical realms. The apparatus itself evolves into a superposition, entangled with the state of the thing being observed. The apparently definite measurement outcome ("the electron is spin-up") is only relative to a particular state of the apparatus ("I measured the electron to be spin-up"). The other possible measurement outcomes still exist and are perfectly real, just as separate worlds. All we have to do is to courageously face up to what quantum mechanics has been trying to tell us all along.

○ ○ ○